

NONABELIAN COMPOSITION FACTORS OF FINITE GROUPS WHOSE MAXIMAL SUBGROUPS ARE COMPLEMENTED

RICHIE SATER

ABSTRACT. Let G be a finite group such that every maximal subgroup M has an exact complement: $G = MK$ and $M \cap K = 1$ for some $K \leq G$. We prove that every nonabelian composition factor of G is isomorphic to

$$\mathrm{PSL}_2(7), \quad \mathrm{PSL}_2(11), \quad \text{or} \quad \mathrm{PSL}_5(2).$$

Conversely, all three occur. This answers Kourovka Problem 18.68.

The main point is to avoid passing the hypothesis to sections, which is false. We first pass to a monolithic quotient with socle S^k . For any almost-simple coordinate group $S \leq X \leq \mathrm{Aut}(S)$, we manufacture a new primitive product action of that quotient. A theorem of Liebeck–Praeger–Saxl forces the coordinate action to have a core-free transitive subgroup. Maximal-factorization classifications exclude this for every simple S outside the displayed list. The only infinite factor-screen survivor, $S = \mathrm{PSp}_4(2^f)$, is eliminated by a 2-adic lifting argument in a prime-degree subfield action. The prime-field subgroup meets every involution class. Fifteen named small almost-simple cases are certified by complete tables of marks; all infinite coverage comes from published exhaustive classifications.

1. INTRODUCTION

Say that a finite group G has property **CMP** if every maximal subgroup $M < G$ has a subgroup $K \leq G$ such that

$$(1) \quad G = MK, \quad M \cap K = 1.$$

The complement is not assumed normal. Kourovka Problem 18.68 asks for the nonabelian composition factors of a nonsoluble group with this property [8, Problem 18.68]. Levchuk and Likharev proved that the nonabelian simple groups having **CMP** are precisely $\mathrm{PSL}_2(7), \mathrm{PSL}_2(11), \mathrm{PSL}_5(2)$ [10, Theorem 1]. The issue is that **CMP** does not pass to normal subgroups or sections. For example, D_8 has **CMP**, while its normal cyclic subgroup C_4 does not.

Our main result is the expected section-safe extension of the simple-group classification.

Date: 11 August 2026.

Theorem 1.1. *Let G be a finite group with CMP. Every nonabelian composition factor of G is isomorphic to one of*

$$\mathrm{PSL}_2(7), \quad \mathrm{PSL}_2(11), \quad \mathrm{PSL}_5(2).$$

Each of the three possibilities occurs.

The proof uses the classification of finite simple groups (CFSG), the maximal-factorization classifications cited in [Section 4](#), and Corollary 3 of Liebeck–Praeger–Saxl [12]. The finite calculation in [Section 7](#) concerns only fifteen explicitly named almost-simple groups. No bounded computation is used to infer an infinite family statement.

2. FROM A COMPOSITION FACTOR TO A MONOLITHIC QUOTIENT

We begin with the inheritance statements that are actually valid.

Lemma 2.1 (quotient closure). *If G has CMP and $N \trianglelefteq G$, then G/N has CMP.*

Proof. Let $\overline{M}$ be maximal in G/N , and let M be its full preimage. Choose $K \leq G$ satisfying (1). Then $G/N = \overline{M}(KN/N)$. If $kN \in \overline{M}$ with $k \in K$, then $k \in M$, and hence $k = 1$. Thus KN/N is an exact complement to $\overline{M}$. $\square$

Lemma 2.2 (regular-action reformulation). *A subgroup $K \leq G$ satisfies (1) if and only if it is regular on the right cosets of M . If M is maximal and core-free, this is a faithful primitive action.*

Proof. Transitivity is equivalent to $G = MK$, and the stabilizer in K of the coset M is $M \cap K$. $\square$

Proposition 2.3 (monolithic reduction). *Let S be a nonabelian composition factor of a finite CMP-group G . There is a quotient L of G such that*

- (1) L has CMP;
- (2) $\mathrm{Soc}(L) = N \cong S^k$ for some $k \geq 1$, and N is the unique minimal normal subgroup of L ;
- (3) $C_L(N) = 1$.

Proof. Choose a chief factor $A/B \cong S^k$ of G , and define

$$C/B = C_{G/B}(A/B), \quad L = G/C, \quad N = AC/C.$$

Since A/B is centerless, $A \cap C = B$, and hence $N \cong A/B$. Chiefness makes N a minimal normal subgroup of L . The conjugation action of L on N is faithful by the definition of C , so $C_L(N) = 1$. If R were another minimal normal subgroup, then $[R, N] \leq R \cap N = 1$, forcing $R \leq C_L(N) = 1$. Thus N is unique. Finally, L has CMP by [Lemma 2.1](#). $\square$

We shall use a stronger fact than the existence of one primitive action: the same abstract group L has many primitive actions, one for each suitable action of its coordinate group.

3. THE PRODUCT-ACTION LIFTING THEOREM

Write

$$N = S_1 \times \cdots \times S_k \cong S^k.$$

The action of L on the factors is transitive, since the product of the factors in any orbit is normal in L . Set

$$(2) \quad X = N_L(S_1)/C_L(S_1),$$

identifying S_1 with its inner automorphism group. Then $S \leq X \leq \text{Aut}(S)$. The standard wreath embedding gives

$$(3) \quad L \hookrightarrow X \wr P \leq X \wr \text{Sym}(k),$$

where P is transitive, N is the base socle S^k , and the group induced on a coordinate is X . This embedding is faithful because $C_L(N) = 1$.

Let $H < X$ be core-free and maximal. Put $\Delta = X/H$ and $V = H \cap S$, and assume $V \neq 1$. Since S is normal and is not contained in H , we have $X = SH$; in particular, S is transitive on Δ . Via (3), let L act on Δ^k in product action.

Lemma 3.1 (primitivity). *The action of L on Δ^k is faithful and primitive.*

Proof. Let $\omega = (H, \dots, H)$, $M = L_\omega$, and $N = S^k$. The group N is transitive, so $L = MN$ and $M \cap N = V^k$.

Suppose $M \leq J \leq L$, and put $D = J \cap N$. Then $J = MD$, $D \trianglelefteq M$, and $V^k \leq D$. The image of M on the coordinates is transitive: multiply an element of L by a suitable base element to make it fix ω without changing its top permutation. Moreover, the subgroup of M stabilizing coordinate i induces H there. The nontrivial inclusion follows by taking an element inducing a prescribed $h \in H$ and correcting all other coordinates with elements of N .

Let D_i be the projection of D to $S_i \cong S$. It contains V and is normalized by H . Maximality of H implies either $D_i = V$ or $HD_i = X$. In the second case

$$D_i = (H \cap S)D_i = (HD_i) \cap S = X \cap S = S.$$

Coordinate transitivity gives the same alternative for every i . If all projections are V , then $D = V^k$ and $J = M$. Otherwise D is subdirect in S^k . By Scott's lemma it is a direct product of full diagonal strips. Since $V^k \leq D$ and $V \neq 1$, every strip has length one, so $D = N$ and $J = L$. Thus M is maximal. Faithfulness follows from core-freeness of H and (3). $\square$

Call (X, Δ) *factor-free* if it has no core-free transitive subgroup; equivalently, there is no core-free $C < X$ with $X = HC$.

Proposition 3.2 (coordinate obstruction). *If $(X, X/H)$ is factor-free, then L does not have CMP.*

Proof. For $k = 1$, we have $L = X$. If a regular subgroup R existed, then R would be transitive and $R \cap H = 1$. Its core in X , if nontrivial, would contain the socle S , as X is almost simple. But then $S \leq R$, contrary to $1 \neq H \cap S \leq H$ and $R \cap H = 1$. Thus R would be core-free, contradicting factor-freeness. Let $k \geq 2$. The action in Lemma 3.1 is a primitive product-action with almost-simple component $(X, X/H)$: indeed, its socle stabilizer is V^k , which is nontrivial and is not subdirect in S^k . Corollary 3(iv) of [12] says that if such a group has a regular subgroup, then its component action has a core-free transitive subgroup. Hence no regular subgroup exists. But if L had CMP, its maximal point stabilizer would have a complement, regular by Lemma 2.2, a contradiction. $\square$

This is the step that removes the unbounded exponent k : it converts the entire product-action problem into one almost-simple factorization screen.

4. FACTOR SCREENS AND PRIME LIFTING

We first state the mechanism for applying maximal-factorization tables.

Lemma 4.1 (factor screen). *Let $1 < V < S$, and assume that the S -conjugacy class of V is invariant under X . Choose a maximal subgroup $H < X$ containing $N_X(V)$. Then H is core-free and $X = HS$. If X/H is not factor-free, there are $S \leq Y \leq X$ and a nontrivial maximal factorization $Y = AB$ such that, after interchanging the factors,*

$$V \leq A \cap S.$$

Proof. Class invariance gives $N_X(V)S = X$, so no proper overgroup of $N_X(V)$ contains S . Hence H is core-free and $HS = X$.

Suppose $C < X$ is core-free and $X = HC$. Put $Y = CS$. Then $Y = (H \cap Y)C$, and both factors supplement S in Y . Consequently all their proper maximal overgroups are core-free. Enlarging them gives a maximal factorization $Y = AB$, with $V \leq H \cap S \leq A \cap S$. $\square$

The remaining tool is the prime-divisibility analogue of the obstruction used in related subgroup-product problems.

Definition 4.2. An action of Z on Λ is *p-elusive* if $p \mid |\Lambda|$ and every element of order p in Z fixes a point.

Lemma 4.3 (wreath lifting). *If X is p-elusive on Δ , then $X \wr \text{Sym}(k)$ is p-elusive on Δ^k . In particular, no subgroup of this wreath product is regular on Δ^k .*

Proof. Let $w = (x_1, \dots, x_k)\sigma$ have order p . On a fixed coordinate of σ , the corresponding x_i has order 1 or p , and hence fixes a point. On a p -cycle of σ , the cycle product of the x_i is 1, since $w^p = 1$; the fixed-tuple equations can therefore be solved successively around the cycle. Thus w fixes a tuple.

If a subgroup were regular, Cauchy's theorem would give it a nonidentity element of order p , because its order is $|\Delta|^k$. That element would fix a point, a contradiction. $\square$

Lemma 4.4 (socle-valuation lifting). *Suppose S is p -elusive on $\Delta = X/H$. Let*

$$a = v_p(|\Delta|), \quad o = v_p(|X/S|).$$

If $a > o$, then no subgroup $L \leq X \wr \text{Sym}(k)$ containing S^k is regular on Δ^k .

Proof. Suppose R is regular and put $Q = R \cap S^k$. Since $R/Q \hookrightarrow L/S^k$,

$$v_p(|R/Q|) \leq ko + v_p(k!).$$

As $v_p(|R|) = ka$,

$$v_p(|Q|) \geq k(a - o) - v_p(k!) > 0,$$

because $a - o \geq 1$ and $v_p(k!) < k$. An element of order p in Q fixes a tuple coordinatewise, contrary to regularity. $\square$

5. THE COORDINATE CLASSIFICATION

Proposition 5.1. *Let S be a nonabelian finite simple group not isomorphic to $\text{PSL}_2(7)$, $\text{PSL}_2(11)$, or $\text{PSL}_5(2)$, and let $S \leq X \leq \text{Aut}(S)$. There is a core-free maximal subgroup $H < X$, with $H \cap S \neq 1$, for which one of the following holds:*

- (1) X/H is factor-free;
- (2) X is p -elusive on X/H for some prime p ;
- (3) $S = \text{PSp}_4(2^f)$, $f \geq 2$, and the hypotheses of [Lemma 4.4](#) hold with $p = 2$.

Proof.

5.1. Alternating groups. For $n \neq 6$, the coordinate group is A_n or S_n . The following actions are p -elusive:

Socle	action and degree	prime
A_5	2-subsets, 10	2
A_6 , $X = A_6$	natural, 6	2
A_6 , $X = S_6$	unordered 3 + 3 partitions, 10	2
A_8	2-subsets, 28	2
A_9	2-subsets, 36	2
A_{10}	4-subsets, 210	2
A_{11}	3-subsets, 165	3
A_{12}	2-subsets, 66	2

The assertion is immediate from cycle supports. An involution fixes the support of one 2-cycle. In degree 10, every involution also has an invariant 4-set: use two 2-cycles, or one 2-cycle and two fixed points. An element of order 3 in degree 11 fixes the support of one 3-cycle. For the S_6 partition action, the cycle types $2, 2^2, 2^3$ respectively preserve a 3-set or interchange it with its complement.

The three outer extensions of A_6 not equal to S_6 , and the two groups A_7, S_7 , have factor-free maximal subgroups certified in [Section 7](#). Their orders and indices are

X	$A_{6.2_2}$	$A_{6.2_3}$	$A_{6.2^2}$	A_7	S_7
$ H $	16	20	32	72	144
$ X : H $	45	36	45	35	35.

Finally let $n \geq 13$, and take the stabilizer of a 6-subset. It is maximal because $6 < n/2$. The alternating-group part of the LPS maximal factorization theorem [[11](#), Theorem D and Remark 2] says that an intransitive factor in a nontrivial maximal factorization stabilizes a j -subset with $j \leq 5$, apart from the listed cases of degrees 6, 8, 10. Thus the 6-set action is factor-free.

5.2. Classical groups. We use Theorem 2.15 and Appendix Tables A.1–A.7 of Xia–Li [[14](#)], a restatement of the LPS classification of nontrivial maximal factorizations with classical socle, together with the two corrected $\text{P}\Omega_8^+(4)$ and $\text{P}\Omega_8^+(16)$ rows of Gill–Giudici–Spiga [[5](#), Section 2]. In the table below, V is invariant up to S -conjugacy under X . Applying [Lemma 4.1](#), inspection of the indicated factorization table shows that no factor intersection contains V .

Socle and conditions	screen subgroup V	table check
$\text{PSL}_2(q)$, excluding the allowed and alternating cases	split-torus normalizer	not in the nonsplit-torus factor of A.1:1 or in P_1 ; A.1:5–8 are checked below
$\text{PSL}_3(q)$, $q \neq 4$, excluding $q = 2$	split-torus normalizer	not in either parabolic or the extension-field factor in A.1:1
$\text{PSL}_3(4)$, every coordinate group X	a pinned maximal subgroup of X	complete Table-of-Marks certificate in Section 7
$\text{PSL}_n(q)$, even $n \geq 4$	$P_{n/2}$	absent from A.1:1–4 (and A.1:5–10 have other dimensions)
$\text{PSL}_n(q)$, odd $n \geq 5$, no graph	P_2	absent from A.1:1–4,10 except $(n, q) = (5, 2)$
$\text{PSL}_n(q)$, odd $n \geq 5$, graph present	$P_{2, n-2}$	only maximal parabolic overgroups P_2, P_{n-2} ; A.1:1–4,10 are excluded except $(5, 2)$
$\text{PSp}_{2m}(q)$, $m \geq 3$	P_2	absent from A.2:1–16 (A.2:13 has P_2 only when $m = 2$)

Socle and conditions	screen subgroup V	table check
$\mathrm{PSp}_4(q)$, q odd	nonsingular $2 + 2$ decomposition stabilizer	absent from the $m = 2$, odd- q rows A.2:1,13
$\mathrm{PSU}_3(q)$	N_1	absent from the $n = 3$ rows A.3:5–7
$\mathrm{PSU}_4(q)$, $q \geq 3$	full nonsingular $2 + 2$ decomposition stabilizer	absent from A.3:1,2,9
$\mathrm{PSU}_n(q)$, $n \geq 5$	N_2	absent from A.3:1–4,10–12
$\Omega_{2m+1}(q)$, $m \geq 3$, q odd	P_2	absent from A.4:1–8
$\mathrm{P}\Omega_{2m}^-(q)$, $m \geq 4$	P_2	absent from A.5:1–5
$\mathrm{P}\Omega_{2m}^+(q)$, $m \geq 5$	P_2	absent from A.6:1–13
$\mathrm{P}\Omega_8^+(q)$, triality allowed	central-node P_2	fixed by triality and absent from A.7:1–15

The two corrected orthogonal factorizations have socle intersections N_2^- and a triality image of $\Omega_8^-(q^{1/2})$, for $q = 4, 16$. Neither contains the central-node maximal parabolic P_2 , so they do not alter the last row of the screen.

Here a standard maximal parabolic can only be contained in a proper factor intersection if it is that intersection. A subgroup containing $P_{2,n-2}$ contains a Borel and is therefore parabolic; its only proper overgroups are P_2 and P_{n-2} . These observations justify the containment, rather than mere equality, assertions in the table. The displayed N_i and full $2 + 2$ decomposition stabilizers are standard maximal geometric subgroups for the stated parameters [9, Chapters 3–4]; the exceptional $\mathrm{PSU}_4(2)$ parameter is routed through the low-rank isomorphism below.

For the first row, the split-torus normalizer has order $2(q-1)/(2, q-1)$ and interchanges the two fixed points of its torus, so it is not contained in P_1 . Nor can it lie in the nonsplit-torus normalizer: its order would force $q-1 \mid q+1$, apart from nonsimple parameters. The other A.1 factors occur only for

$$q = 7, 11, 16, 19, 23, 29, 59.$$

The values 7, 11 are allowed. At $q = 19, 29, 59, 23$, divisibility by the torus-normalizer order excludes A_5 or S_4 . At $q = 16$, a subgroup of order 30 lies neither in D_{34} nor in A_5 , the latter because it would have index 2. This closes the rank-one family.

For $S = \mathrm{PSL}_3(q)$, let $d = (3, q-1)$ and let V normalize a maximally split torus. Its class is invariant under diagonal, field, and graph automorphisms.

The monomial action shows that V fixes neither a point nor a hyperplane, and

$$|V| = \frac{6(q-1)^2}{d}.$$

The extension-field factor in row 1 of Table A.1 has intersection with S of order $3(q^2 + q + 1)/d$. For $q = 3$, divisibility excludes containment; for $q \geq 5$, the former order is larger. The only remaining table row is the exceptional $q = 4$ row, and all ten almost-simple groups with that socle are handled by the finite certificate. The parameter $q = 2$ gives the allowed group $\mathrm{PSL}_3(2) \cong \mathrm{PSL}_2(7)$.

It remains to treat $S = \mathrm{PSp}_4(q) = \mathrm{Sp}_4(q)$ with $q = 2^f$, $f \geq 2$. Choose a prime $r \mid f$, put $d = f/r$, $q_0 = 2^d$, and let

$$V = \mathrm{Sp}_4(q_0) < S$$

be the standard subfield subgroup. It is maximal in S , since the field-extension degree r is prime [1, Table 8.14]; see also [2, Proposition 4.2 and Table 3].

We check explicitly that this one choice works for every field-only and graph-outer coordinate group $S \leq X \leq \mathrm{Aut}(S)$. Realize S as the fixed points of the standard Frobenius φ^f in the adjoint algebraic group of type B_2 over $\overline{\mathbb{F}}_2$. The exceptional graph-field endomorphism ρ satisfies $\rho^2 = \varphi$, and $\mathrm{Aut}(S) = \langle S, \rho \rangle$ [7, Lemma 2.1]. Since ρ commutes with φ , it normalizes the fixed-point subgroup $V = \mathrm{Sp}_4(2^d)$. Thus the S -class of V is invariant under all of $\mathrm{Aut}(S)$.

Put $H = N_X(V)$. Class invariance gives $X = SH$, and $H \cap S = N_S(V) = V$. Moreover, H is maximal in X . Indeed, if $H < K < X$, then maximality of V in S gives $K \cap S = V$ or S . In the first case, $V = K \cap S \trianglelefteq K$, so $K \leq N_X(V) = H$; in the second, $S \leq K$ and $X = SH \leq K$. Both are contradictions. Finally, H is core-free: every nontrivial normal subgroup of the almost-simple group X contains S , whereas $H \cap S = V < S$.

Every involution class of S meets V . Here are the details needed at the prime-field endpoint $q_0 = 2$. By Lemmas 2.1–2.2 and Corollary 2.3 of [13], the three involution classes of S meet

$$O_4^+(q) \cong (\mathrm{SL}_2(q) \times \mathrm{SL}_2(q)) \rtimes \langle \tau \rangle,$$

where τ interchanges the factors, and conjugacy inside this orthogonal group agrees with conjugacy in S . If u represents the unique nonidentity involution class in $\mathrm{SL}_2(q)$, the three orthogonal classes have representatives $(u, 1), (u, u), \tau$. We may take $u \in \mathrm{SL}_2(2)$, so all three representatives lie in $O_4^+(2) \leq \mathrm{Sp}_4(2) \leq V$. Hence S is 2-elusive on X/H .

Since $X = SH$, the degree and its 2-part are

$$(4) \quad |X : H| = |S : V| = \frac{q^4(q^2 - 1)(q^4 - 1)}{q_0^4(q_0^2 - 1)(q_0^4 - 1)}, \quad a = v_2(|X : H|) = 4(f - d).$$

The same outer-automorphism description gives $|X/S| \mid 2f$, so

$$o = v_2(|X/S|) \leq 1 + v_2(f) < 2f \leq 4(f - d) = a.$$

Here $v_2(f) \leq f - 1$, and $f - d = f(1 - 1/r) \geq f/2$. Thus [Lemma 4.4](#) applies uniformly. In particular, when $q = 4$ and $X = \text{Aut}(S)$, the corrected subgroup has $|H| = 2880$, $|H \cap S| = 720$, and degree 1360; these values are also reproduced in [Section 7](#). This is the sole infinite classical row for which the factor screen itself does not suffice.

The standard low-rank isomorphisms

$$\begin{aligned} \text{PSL}_2(4) &\cong \text{PSL}_2(5) \cong A_5, & \text{PSL}_2(9) &\cong A_6, \\ \text{PSL}_3(2) &\cong \text{PSL}_2(7), & \text{PSL}_4(2) &\cong A_8, \\ \text{PSU}_4(2) &\cong \text{PSp}_4(3). \end{aligned}$$

and the $B_2 = C_2$, $D_3 = A_3$, ${}^2D_3 = {}^2A_3$ identifications route every overlap to a row already considered.

5.3. Exceptional groups of Lie type. Hering–Liebeck–Saxl determine all factorizations of exceptional simple groups and their automorphism groups [\[6, Theorems 1–2\]](#). The only socles admitting a core-free factorization are

$$G_2(4), \quad G_2(3^f), \quad F_4(2^f),$$

together with one extra outer factorization for $G_2(4).2$. None of the listed factors is parabolic. For every exceptional socle outside these three families, take V to be a Sylow subgroup in the defining characteristic. Its S -conjugacy class is invariant under every automorphism, and the Hering–Liebeck–Saxl theorem and [Lemma 4.1](#) make the resulting action factor-free.

It remains to screen the three factorable families. If no graph automorphism is present, choose a maximal parabolic. In the graph-fused cases choose the graph-stable parabolic intersection $P_1 \cap P_2$ in $G_2(3^f)$, or $P_1 \cap P_4$ in $F_4(2^f)$; each contains a common Borel. Their respective p -parts are q^6 and at least q^{24} . The p -parts of the displayed Hering–Liebeck–Saxl factors are at most q^3 and q^{16} , so no factor contains the chosen screen subgroup. The small isomorphisms $G_2(2)' \cong \text{PSU}_3(3)$ and ${}^2G_2(3)' \cong \text{PSL}_2(8)$ have already been covered.

5.4. Sporadic groups. Giudici classifies all factorizations with sporadic socle [\[4, Theorems 1.1–1.2 and Tables 1–4\]](#). Comparing those tables with the Atlas maximal-subgroup lists [\[3\]](#) gives the following factor-free maximal subgroups. A row for an outer extension also excludes the factorizations lifted from the simple socle.

X	H	X	H
M_{11}	S_5	M_{12}	$2^3:S_4$
$M_{12}.2$	$3^{1+2}:D_8$	$M_{22}.2$	$A_6.2^2$
M_{23}	A_8	J_2	$3.A_6.2_2$
$J_2.2$	$3.A_6.2^2$	HS	$L_3(4).2_1$
$HS.2$	$L_3(4).2^2$	He	$2^2.L_3(4).S_3$
$He.2$	$2^2.L_3(4).D_{12}$	Ru	$2^6.U_3(3):2$
Suz	$J_2.2$	$Suz.2$	$J_2.2 \times 2$
Fi_{22}	$O_7(3)$	$Fi_{22}.2$	$G_2(3).2$
Co_1	$2^{11}:M_{24}$		

The simple group M_{22} , and every sporadic almost-simple group not needed in this display, has no nontrivial core-free factorization. For such a group, apply [Lemma 4.1](#) to a Sylow subgroup of the socle; its conjugacy class is automorphism-invariant and its maximal overgroup has nontrivial socle intersection. The rows available in TomLib are independently checked in [Section 7](#).

The sole factor-cover survivor is M_{24} . Let it act on the 2-subsets of its natural 24-point set. The point stabilizer $M_{22}.2$ is maximal, the degree is 276, and every involution fixes the support of one of its 2-cycles. Thus the action is 2-elusive and [Lemma 4.3](#) applies.

This completes the CFSG audit. $\square$

6. PROOF OF THE MAIN THEOREM

Proof of [Theorem 1.1](#). Let S be a nonabelian composition factor of G . Apply [Proposition 2.3](#) to obtain a CMP-quotient L with unique self-centralizing socle S^k , and define its coordinate group X by [\(2\)](#).

Suppose S is not one of the three groups in the conclusion. Choose the maximal subgroup $H < X$ supplied by [Proposition 5.1](#). By [Lemma 3.1](#), the same group L has a faithful primitive action on $(X/H)^k$. If the action X/H is factor-free, [Proposition 3.2](#) contradicts $\text{CMP}(L)$. If it is p -elusive, [Lemma 4.3](#) shows that the primitive action has no regular subgroup, again contradicting $\text{CMP}(L)$. In the final symplectic case, the same contradiction follows from [Lemma 4.4](#). Therefore S is one of the three displayed groups.

Conversely, Levchuk–Likharev prove that each of the three simple groups has CMP [[10](#), Theorem 1]; taking the group itself shows that each occurs as a composition factor. $\square$

7. FINITE TABLE-OF-MARKS CERTIFICATE

The proof uses finite certificates for five small alternating coordinate groups and for the ten almost-simple coordinate groups with socle $\text{PSL}_3(4)$. Nine sporadic rows are also cross-checked.

For a complete table of marks T , let h, c, i represent conjugacy classes of subgroups H, C, I . The command `IntersectionsTom(T,h,c)[i]` is positive

exactly when suitable conjugates of H, C have intersection in class i . Such a pair factors X precisely when

$$|H||C| = |X||I|.$$

A subgroup C is core-free in an almost-simple X precisely when its class does not contain the socle class. The producer checks every core-free class c and every intersection class i , verifies $|H \cap S| = |H||S|/|X| > 1$, and fails on missing data.

The proof rows are:

X	$A_6.2_2$	$A_6.2_3$	$A_6.2^2$	A_7	S_7
$ H $	16	20	32	72	144
$ X : H $	45	36	45	35	35.

Since $\text{Out}(\text{PSL}_3(4)) \cong D_{12}$, the following ten TomLib names exhaust, up to isomorphism, the groups between the socle and its full automorphism group:

X	$ H $	$ X:H $	X	$ H $	$ X:H $
$L_3(4)$	960	21	$L_3(4).2_1$	384	105
$L_3(4).2_2$	1920	21	$L_3(4).2_3$	720	56
$L_3(4).3$	216	280	$L_3(4).2^2$	768	105
$L_3(4).3.2_2$	432	280	$L_3(4).3.2_3$	1152	105
$L_3(4).6$	1152	105	$L_3(4).D_{12}$	2304	105.

Every row has zero core-free factor classes. The producer and independent checker are, respectively,

```
gap --quitionbreak -q gap/generate-factor-free-scan.g
python3 tests/check-factor-free-scan.py
```

in the accompanying repository. The environment is GAP 4.15.1 with TomLib 1.2.11. A separate GAP 4.15.1/AtlasRep 2.1.11 regression test reconstructs the corrected $S_4(4).4$ row just quoted. All GAP commands use `-quitionbreak`; the Python checkers are also run under `python3 -O`, so their acceptance checks do not depend on assertions. The generated TSV has SHA-256

9b131720d41ef945a0696794c0493ae9c07e166d4d1f37b054a1d55f0c4837ae

and records all twenty-four proof and cross-check rows. This calculation is an exact finite verification relative to the complete named tables, not an extrapolation from a library scan.

8. DEPENDENCY AND SCOPE STATEMENT

The universal conclusion depends on CFSG and on the exhaustive published factorization classifications [11, 14, 5, 6, 4]. The passage from S^k to one coordinate uses [12, Corollary 3]. The subfield maximality and outer-automorphism inputs for $\text{P}\text{Sp}_4(2^f)$ are [1, Table 8.14] and [7, Lemma 2.1]; the involution-class input is [13, Lemmas 2.1–2.2 and Corollary 2.3]. No property of a section is inferred from CMP; only quotient closure is used.

The July 2026 Kourovka Notebook still prints Problem 18.68 without a solution annotation. That dated bibliographic fact is not needed in the proof and is not a claim that no unindexed or contemporaneous solution exists.

AI-USE DISCLOSURE

AI tools were used extensively for literature search, proof exploration, code generation, drafting, and adversarial review. The named author is solely responsible for verifying the arguments, sources, computations, and final submitted text.

REFERENCES

- [1] J. N. Bray, D. F. Holt, and C. M. Roney-Dougal, *The Maximal Subgroups of the Low-Dimensional Finite Classical Groups*, London Math. Soc. Lecture Note Ser. **407**, Cambridge University Press, Cambridge, 2013, <https://doi.org/10.1017/CB09781139192576>.
- [2] T. C. Burness, On base sizes for actions of finite classical groups, *J. London Math. Soc.* (2) **75** (2007), 545–562, <https://doi.org/10.1112/jlms/jdm033>.
- [3] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker, and R. A. Wilson, *Atlas of Finite Groups*, Clarendon Press, Oxford, 1985.
- [4] M. Giudici, Factorisations of sporadic simple groups, *J. Algebra* **304** (2006), 311–323, <https://doi.org/10.1016/j.jalgebra.2006.04.019>.
- [5] N. Gill, M. Giudici, and P. Spiga, A generalization of Szep’s conjecture for almost simple groups, *Vietnam J. Math.* **52** (2024), 325–359, <https://doi.org/10.1007/s10013-023-00635-1>.
- [6] C. Hering, M. W. Liebeck, and J. Saxl, The factorizations of the finite exceptional groups of Lie type, *J. Algebra* **106** (1987), 517–527, [https://doi.org/10.1016/0021-8693\(87\)90013-5](https://doi.org/10.1016/0021-8693(87)90013-5).
- [7] S. Harper, Totally deranged elements of almost simple groups and invariable generating sets, *J. London Math. Soc.* (2) **109** (2024), no. 6, e12935, <https://doi.org/10.1112/jlms.12935>.
- [8] E. I. Khukhro and V. D. Mazurov, eds., *The Kourovka Notebook: Unsolved Problems in Group Theory*, 21st ed., 2026, <https://alglog.org/21tkr.pdf>.
- [9] P. B. Kleidman and M. W. Liebeck, *The Subgroup Structure of the Finite Classical Groups*, London Math. Soc. Lecture Note Ser. **129**, Cambridge University Press, Cambridge, 1990, <https://doi.org/10.1017/CB09780511629235>.
- [10] V. M. Levchuk and A. G. Likharev, Finite simple groups with complemented maximal subgroups, *Siberian Math. J.* **47** (2006), 659–668, <https://doi.org/10.1007/s11202-006-0077-7>.
- [11] M. W. Liebeck, C. E. Praeger, and J. Saxl, *The Maximal Factorizations of the Finite Simple Groups and Their Automorphism Groups*, Mem. Amer. Math. Soc. **86** (1990), no. 432, <https://bookstore.ams.org/memo-86-432>.
- [12] M. W. Liebeck, C. E. Praeger, and J. Saxl, Transitive subgroups of primitive permutation groups, *J. Algebra* **234** (2000), 291–361, <https://doi.org/10.1006/jabr.2000.8547>.
- [13] M. W. Liebeck, C. E. Praeger, and J. Saxl, Regular subgroups of primitive permutation groups, *Mem. Amer. Math. Soc.* **203** (2010), no. 952, <https://doi.org/10.1090/S0065-9266-09-00569-9>.
- [14] C. H. Li and B. Xia, *Factorizations of Almost Simple Groups with a Solvable Factor, and Cayley Graphs of Solvable Groups*, Mem. Amer. Math. Soc. **279** (2022), no. 1375, <https://doi.org/10.1090/memo/1375>.